

Varshit Manepalli

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Experience

Machine Learning Intern, Escape Enterprises – Indianapolis, IN Dec 2024 – Present

- Built and deployed ML models and APIs for mood tracking, body language analysis, and environment detection
- Integrated speech-to-text and text-to-speech features to custom built LLM Lucille API, a conversational AI
- Trained a Bidirectional LSTM model on pose keypoints from video data (RAVDSS) for body language analysis
- Developed a recommendation engine using user mood history and feedback for personalized self-care services
- Collaborated with UX and clinical teams to design, test, and improve features across web and mobile platforms

Machine Learning Engineer, Progress Solutions – Plano, TX Apr 2024 – Nov 2024

- Architected scalable database architectures, reducing query response time by 60% during peak traffic
- Designed and deployed Flask modules for machine learning models, increasing deployment efficiency by 35%
- Engineered a BERT-based hate speech detection system, minimizing false positives by 15%
- Integrated LLMs to create conversational AI systems, boosting user engagement by 25%
- Conducted EDA to address class imbalances and collaborated on data warehouse design

Software Developer Intern, Divami Design Labs – Hyderabad, India Jan 2022 – Jul 2022

- Directed mobile-responsive web design projects, boosting mobile engagement by 60% and site retention by 30%
- Developed centralized employee and healthcare management systems, streamlining HR operations
- Identified and resolved hardware-software integration bottlenecks, improving system efficiency by 25%
- Designed RESTful APIs and optimized software testing, reducing release cycles by 15% and downtime by 20%

Skills

Programming Languages: Python, SQL, R, Java, MATLAB, JavaScript, HTML, CSS, PHP, JSON

Frameworks & Libraries: Flask, Django, PyTorch, TensorFlow, NumPy, Pandas, Keras, PySpark, scikit-learn

Platforms & APIs: Google Cloud VERTEX AI, Restful APIs, ETL applications, CI/CD frameworks

Projects

Cyberattack Pattern Detection and Prevention with GANs

- Developed a GAN-based threat simulation system, generating over 200k cyberattack scenarios, merging synthetic and real-world datasets, and automating workflows to reduce manual intervention by 30%
- **Tools Used:** PyTorch, TensorFlow, NVIDIA AI Workbench

Hate Speech Detection Using LLMs

- Built a hate speech detection pipeline using advanced language models, featuring a REST API with 200ms response time, improved recall for minority classes by 15%, and ensured robustness across multilingual datasets
- **Tools Used:** Python, TensorFlow, BERT, GPT, Flask

Innovative HIV Therapy Research Using Real-World Data

- Optimized HIV therapy trial designs using real-world clinical data, leveraging propensity score matching, predictive models to improve patient outcomes by 20%, and automated data pre-processing
- **Tools Used:** Python, Excel, Power Query

Education

Stevens Institute of Technology, MS in Computer Science Sep 2022 – May 2024

- **Coursework:** Java Programming, Data mining, Data Analytics, Machine Learning, Deep learning

VIT-AP University, B.tech in Computer Science and engineering Aug 2018 – Jun 2022

- **Coursework:** Data Structures & Algorithms, Calculus, Differential Equations, Applied Statistics